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Decentralized Certification on Bitcoin

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Introduction

- Verification of a document is essential for maintaining organizational trust
- The centralization of verification incurs significant expenses
- Bitcoin is the first decentralized network that is: permissionless, highly tolerant to DDoS attacks, and based on Proof-of-Work (PoW)

Motivation

- Legacy Bitcoin solutions are not scalable
- **Using alternative networks such as Ethereum introduces different security and trust considerations**, increasing consensus complexity and **censorship risks**.
- Major Bitcoin upgrades (SegWit and Taproot) enable new possibilities on the base layer (**L1**) by improving **script efficiency** and **privacy**, without relying on external layers.

Literature Review

Bitcoin is a descentralized `Network`

1. Research Questions

- How can the tools applied by **Taproot** be leveraged to deploy a decentralized verification protocol?
- To what extent does using **MAST** improve **privacy**, **scalability**, and **spatial/computational complexity**?
- What is an appropriate methodology to add signature **key rotation** and **certificate revocation** to the protocol?

2. Specific Objectives

- Design the **Merkle structure of scripts** for verifying the integrity of issued documents.
- Design software components for **ownership verification** (digital signature) and **document signature key rotation**.
- Design a software component that tracks **revocation** to verify the validity of issued documents.
- Deploy a **Proof of Concept (PoC)** on a Bitcoin test network.
- **Measure** the **scalability**, **time/space complexity**, and network **costs (vBytes)** of the proposed protocol.

Protocol Structure and Verification Flow

Protocol Overview (Three Phases)

- **Phase I: Design** ($\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ Objectives) $\rightarrow$ Protocol architecture based on Taproot/MAST.
- **Phase II: Prototyping** $\rightarrow$ $\mathbf{PoC}$ deployment on Testnet/Signet using Python.
- **Phase III: Validation** $\rightarrow$ Experimental analysis of $\mathbf{Scalability}$ and $\mathbf{vByte\ Cost}$.

Core Roles and Components

- **Issuer (Organization):** Provides the document and a '$\mathbf{Proof\ of\ Validation}$' file to the user.
- **Verifier (User):** Requires only the document, the proof, and access to the $\mathbf{public\ Bitcoin\ history}$.
- **Full Verification** requires three successful checks: $\mathbf{Integrity}$, $\mathbf{Ownership}$, and $\mathbf{Revocation}$ status.

Conclusions

Bitcoin is a descentralized `Network`

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